

Nitrogen Cycle in the Earth Climatic System and Its Modeling

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Abstract—Investigations into the nitrogen cycle in the climatic system of the earth are reviewed with special emphasis on the biospheric nitrogen cycle. Approaches to modeling the biogeochemical nitrogen turnover are described. Excluding the nitrogen cycle from consideration when probable consequences of climate change are analyzed can lead to inaccurate estimates of the ecosystem response, in particular, for regions where mineral compounds of soil nitrogen are a limiting factor for the development of vegetation cover. Numerical experiments with climatic models point to a substantial influence of the nitrogen turnover on the feedback between climatic characteristics and the carbon cycle. Models of the combined dynamics of carbon and nitrogen make it possible to obtain realistic estimates of present-day resources and fluxes of these elements in ecosystems, as well as to estimate their changes during possible climatic changes.

Keywords: nitrogen cycle, biosphere, climatic models

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INTRODUCTION

The atmosphere of the earth consists mainly of gaseous compounds of nitrogen. Nitrogen is one of the elements that separated in the gas phase at early stages of the earth's development as a result of shock–explosive processes at the surface. The atmosphere of the early earth developed due to the upper mantle degassing. Nitrogen and ammonia were among gases supplied from the mantle into the atmosphere. Throughout the geological history of the earth, molecular nitrogen and its other gaseous compounds were supplied from the interior of the earth into the atmosphere in the composition of volcanic gases and with waters of thermal sources [1–5].

Changes in the nitrogen content in the earth's atmosphere at different stages of its evolution are presented in Fig. 1 from works [1–4]. According to the data of these investigations, the nitrogen mass permanently increased in the atmosphere in the course of its evolution. Nitrogen most intensely accumulated in the atmosphere in the period of active degassing of the mantle (from 4.25 to 2 Ga ago). It follows from the model calculations [4] that, in the time interval from 2 Ga to 500 Ma ago, the fraction of nitrogen in the atmosphere changed weakly. An insignificant (no more than 4%) increase in the nitrogen mass in the atmosphere over this period was accompanied by the oxygen concentration growth in it. A significant increase in the oxygen concentration in the atmosphere due to the photosynthetic activity of vegetation, which intensely developed on continents, is characteristic of the period 500–400 Ma ago. This process sharply decreased the nitrogen fraction in the

atmosphere. In the subsequent period (from 400 Ma ago to the current epoch), the nitrogen mass in the atmosphere increased insignificantly (by fractions of a percent) with the triple increase in the oxygen mass in the atmosphere.

According to the existing estimates, about 4.5×10^6 Gt of nitrogen were degassed from the mantle into the atmosphere throughout the geological history of the earth [5]. Nitrogen is also released into the atmosphere during rock weathering. The amount of nitrogen released due to rock weathering over the period of the geological history of the earth is about 1% of its mass in the present-day atmosphere [3, 5]. These estimates indicate that the interior of the earth is the main supplier of nitrogen into the atmosphere. The present-day nitrogen content in the atmosphere is estimated at 3.9×10^6 GtN [5, 6].

Gaseous molecular nitrogen, due to its chemical inertness, is the most stable form of the existence of this element. Owing to this, molecular nitrogen gradually accumulated in the atmosphere and has become its main component. The primary nitrogen turnover was abiogenic and was reduced to the photochemical transformation of gaseous nitrogen compounds (apart from its molecular state) into oxidized soluble forms with the subsequent removal from the atmosphere [5]. At the beginning of the geological history of the earth, bacteria were included in the process of nitrogen mass exchange, and these bacteria gradually changed the entire structure of the global nitrogen cycle [6]. Photochemical reactions continue to participate in the removal of nitrogen from the atmosphere; however,

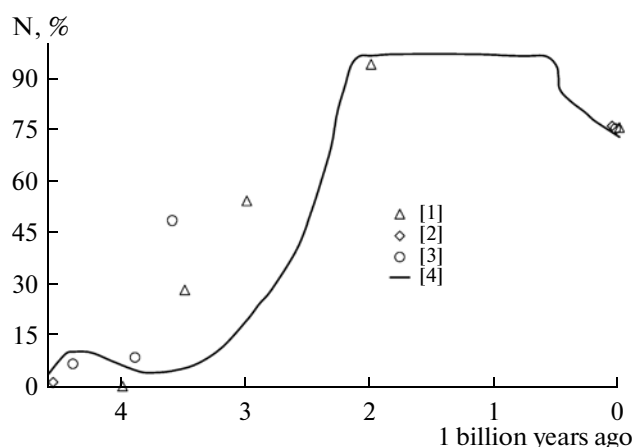


Fig. 1. Changes in the nitrogen (N) content in the earth atmosphere after works [1–4]. The data are in fractions of the atmospheric mass.

the activity of microorganisms is predominant in this process.

Most chemical elements, being transformed as a result of complex biochemical processes from some compounds to others, circulate in the biosphere, forming turnovers that are closed to one or another degree [7]. The biogeochemical cycles of chemical elements are interdependent and interrelated. In this work we emphasize the biospheric nitrogen turnover, which is one of the vital elements of the biosphere. The deficit of nitrogen compounds accessible for plants and microorganisms in the soil adversely affects the intensity of productive and destructive processes in the biosphere. Over the past few decades, due to the increasing amounts of burned fossil fuels, the deforestation of forests, the intensification of agricultural production, and the wider usage of industrial nitrogen-containing fertilizers, the emission of nitrogen oxide into the atmosphere and the accumulation of nitrates in bodies of water have considerably increased, which has a substantial influence on global changes in the environment and climate. The main questions associated with analyzing the biogeochemical nitrogen turnover and its modeling are considered in this review.

Table 1. Distribution of nitrogen over the earth [5, 15, 16]

Component	MtN value
Lithosphere	$5.5 \times 10^8 - 7.7 \times 10^8$
Atmosphere	$3.8 \times 10^9 - 4.0 \times 10^9$
Hydrosphere	$2.0 \times 10^7 - 2.2 \times 10^7$
Terrestrial ecosystems	$1.4 \times 10^5 - 8.2 \times 10^5$

BIOSPHERIC NITROGEN CYCLE AND ITS ROLE IN THE CLIMATIC SYSTEM

The bulk of nitrogen mass on the earth (84.4%) is concentrated in the atmosphere (Table 1). The lithosphere contains about 15.2% of the total nitrogen resources. About 0.4% and 0.002% of the total planetary nitrogen resources are accumulated in the hydrosphere and terrestrial ecosystems, respectively. The nitrogen turnover is one of the main biogeochemical cycles in the natural medium.

Although nitrogen amounts to about 78% of the atmospheric volume, this element is a limiting factor for many physiological processes, which is due to the fact that molecular nitrogen prevails in the atmosphere of the earth. Molecular nitrogen is a chemically stable compound, and it is not assimilated by photosynthesizing plants or the majority of microorganisms. The deficit of mineral forms of soil nitrogen has a negative influence on the development of vegetation cover; namely, this leads to the suppression of above-ground plants, thereby restraining the absorption of carbon by terrestrial ecosystems and favoring global warming [8–10]. The nitrogen cycle affects microbiological processes in the soil; in particular, the deficit of nitrogen in the soil makes the process of detritus (dead organic substance) decomposition slower, which decreases the emission of greenhouse gases as a result of soil respiration [11]. Gaseous nitrogen oxides, which are related to the minor gaseous constituents of the atmosphere, influence the process of absorption of thermal radiation in the atmosphere, enhancing the greenhouse effect [12]. The schematic turnover of nitrogen in the biosphere is shown in Fig. 2.

In the atmosphere, nitrogen is in the molecular state (N_2) and in the form of such compounds as nitrous oxide (N_2O), ammonia (NH_3), nitrogen oxide (NO), nitrogen dioxide (NO_2), and other nitrogen oxides. Chemically inert molecular nitrogen forms the bulk of the atmospheric mass (Table 2). Ammonia plays an important role in the atmosphere, neutralizing sulfur dioxide. As a result of the interaction of ammonia with sulfur dioxide, the concentration of acid-forming substances in the atmosphere decreases, thereby decreasing the acidity of precipitation; in addition, this interaction leads to the formation of ammonium sulfide, which is washed out of the atmosphere by precipitation and serves as a fertilizer for the vegetation of terrestrial ecosystems.

High-temperature reactions, which take place during fires, electric discharges in the atmosphere, and the work of internal-combustion engines, are sources of nitrogen oxides. Gaseous compounds of nitrogen oxides play the key role in the tropospheric photochemical cycle, because they serve as catalysts of oxidizing processes. Higher nitrogen oxides are washed out of the atmosphere by precipitation.

Nitrous oxide is a relatively inert compound. The N_2O concentration in the atmosphere is smaller than

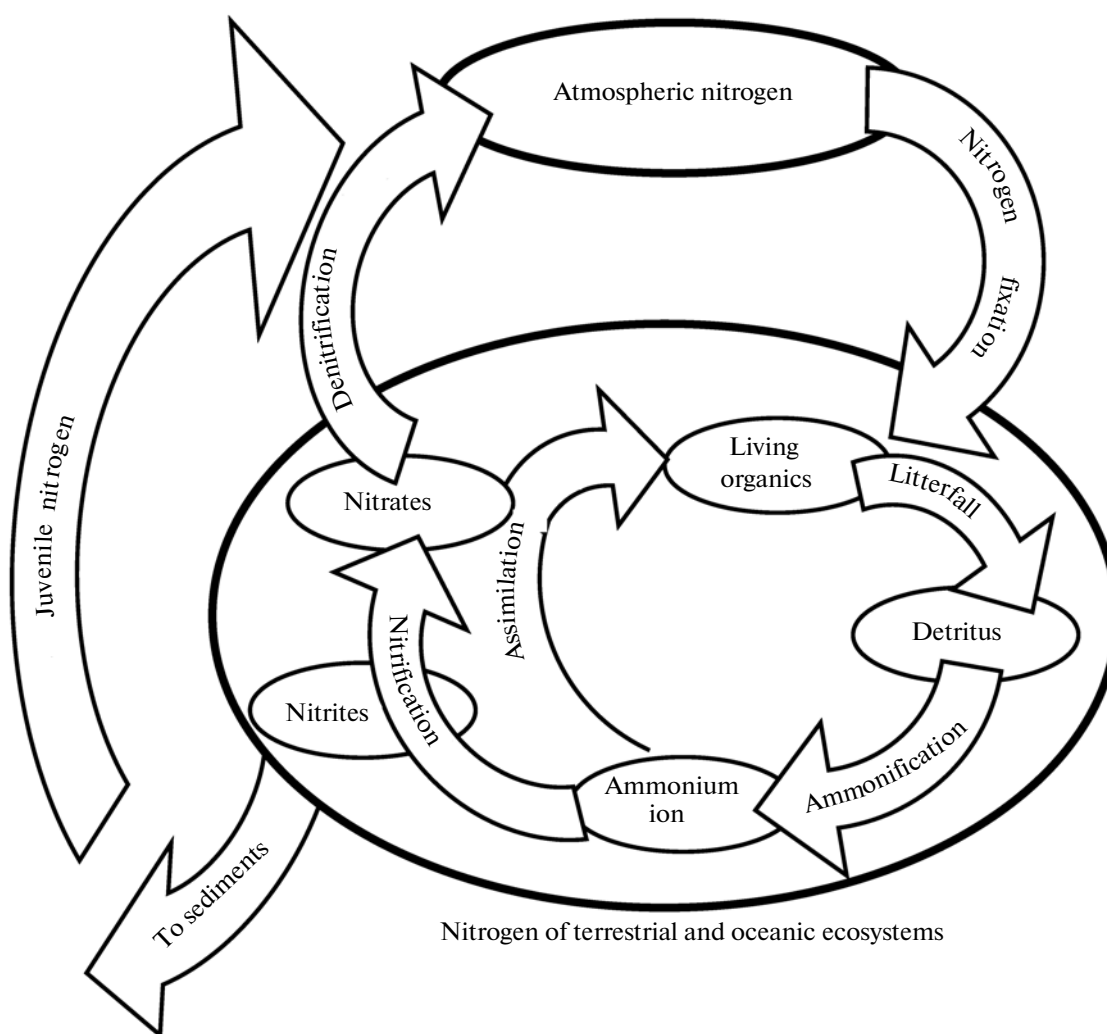


Fig. 2. Scheme of the global nitrogen cycle.

such radiation-active gases as methane and carbon dioxide. However, due to the considerable dwelling time in the atmosphere (on average, more than 100 years) and a substantial greenhouse potential per one molecule (exceeding the potential of carbon dioxide by a factor of 310) [12], nitrous oxide plays an important role in the formation of the greenhouse effect. The contribution of nitrous oxide to the greenhouse effect is estimated at 5–6% [13]. The washout of N_2O by atmospheric precipitation is insignificant compared with higher nitrogen oxides. In the troposphere, nitrous oxide does not participate in photochemical reactions and the concentration of this gas changes little, which points to the absence of tropospheric sink of nitrous oxide. The transport of N_2O from the troposphere into the stratosphere is estimated at 10–

15 Mt/year [6]. In the stratosphere, nitrous oxide is subjected to photolysis and interacts with excited oxygen atoms to form nitrogen oxide, which takes part in photochemical reactions that decrease the ozone concentration.

During volcano eruptions, emissions of fumaroles and geysers, and gas releases in regions of tectonic faults of the crust, juvenile (abyssal) nitrogen is supplied into the atmosphere. Volcanic eruptions noticeably contribute to the total nitrogen turnover: according to the data presented in [15], gases of lavas of Hawaiian volcanoes contain about 6% molecular nitrogen. The high nitrogen content is noted in gases of thermal sources. On average, the annual supply of juvenile nitrogen into the atmosphere is estimated at 1.0 MtN [5, 16].

Table 2. Estimates of nitrogen resources in components of the biosphere [5, 9, 14, 15, 17, 20, 23, 27]

Components	MtN value
Atmosphere:	
N ₂	$3.8 \times 10^9 - 4.0 \times 10^9$
N ₂ O	1100–1800
NH ₃	0.04–0.9
NH ₄ ⁺	0.29–1.8
NO _x	1–4
NO ₃ ⁻	0.16–0.5
organic compounds	1
Terrestrial ecosystems:	
biomass of plants	$10 \times 10^3 - 16 \times 10^3$
biomass of animals	200
supersoil layer of detritus	1900–3300
soils:	
organic matter	$70 \times 10^3 - 820 \times 10^3$
inorganic compounds	$16 \times 10^3 - 140 \times 10^3$
microorganisms	500
Ocean:	
biomass of photosynthesizing organisms	200–800
biomass of animals	170
detritus	$45 \times 10^3 - 900 \times 10^3$
inorganic compounds of nitrogen	$99 \times 10^3 - 690 \times 10^3$
dissolved N ₂	$10 \times 10^6 - 22 \times 10^6$
Earth crust	
rocks	1.7×10^8
sediments	$4 \times 10^8 - 6 \times 10^8$
fossil fuel	1.2×10^5

NO_x = NO + NO₂.

As a result of physicochemical processes in the atmosphere, molecular nitrogen forms its compounds, which are supplied to the land and ocean surfaces with precipitation or as a result of gravitational deposition (*atmospheric nitrogen fixation*). The flux of atmospheric nitrogen fixation depends substantially on the intensity of lightning and the amount of precipitation. The intensity of atmospheric nitrogen fixation is estimated in the range from 10 to 30 MtN/year [17]. In some works this flux is estimated from 80–100 MtN/year [18] to 200 MtN/year [19].

As a result of atmospheric chemical reactions, natural emissions, and anthropogenic discharges, the atmosphere contains nitrate (NO₃⁻) and ammonium (NH₄⁺) aerosols. The global annual contents of these aerosols in the atmosphere are estimated at 0.16–0.48

and 0.29–0.54 MtN, respectively [20, 21]. According to [20], 43% of nitrate and 92% of ammonium aerosols are in the submicron state. Nitrate aerosol increases the ability of the atmosphere to reflect solar radiation, thereby favoring cooling of the surface of the earth. The annual mean global estimates for the anthropogenic direct radiation action of nitrate aerosol at the upper boundary of the atmosphere are in the range from –0.04 to –19 W/m² [12, 21, 22]. Up to 8.6 MtN of nitrate aerosols and 23 MtN of ammonium aerosol are washed by precipitation out of the atmosphere. As a result of gravitational deposition, about 3 MtN/year of nitrate aerosol and 2.8 MtN/year of ammonium aerosol are supplied into terrestrial ecosystems and to the ocean surface [20].

The nitrogen resources in terrestrial ecosystems consist of nitrogen in vegetation (1–2%), in the biomass of animals and microorganisms (less than 0.1%), in the detritus of the supersoil layer (about 0.2%), in mineral forms of soil nitrogen (1–14%), and in organic nitrogen-containing compounds of the soil (7–84%) [15, 23, 24]. Mineral forms of soil nitrogen, amounting to 1–5% of the total nitrogen resources in the soil, are presented by nitrates, nitrites, ammonium salts, and ammonia. Nitrogen contained in these compounds can be assimilated by root systems of plants and by microorganisms. Organic nitrogen-containing compounds in the soil, amounting to 95–99% of the total resources, are presented by complex organic compounds. Organic compounds of soil nitrogen cannot be utilized by higher plants.

Chemically inactive molecular nitrogen of the atmosphere is involved in the process of turnover by some species of microorganisms and plants, which are capable of activating N₂ and binding it to chemical compounds (*biological nitrogen fixation*). Microorganisms capable of binding molecular nitrogen into nitrous compounds exist both in symbiosis with plants and separately. Free-living nitrogen-fixing bacteria bind annually from 3 to 10 gN/m² [25], whereas bacteria existing in symbiosis with plants can accumulate from 15 to 60 gN/m² [26]. The annual accumulation of nitrogen by blue-green algae on irrigated rice fields amounts to 20 gN/m² [15]. On the whole, the intensity of biological nitrogen fixation in natural terrestrial ecosystems is estimated from 30–40 to 200 MtN/year [5, 15].

The nitrogen absorption from soil by plants through their root systems is of a substantial significance in nitrogen turnover. This flux is estimated at 1.2–3.5 GtN/year [5, 17]. The rate of nitrogen assimilation by plant roots depends on the temperature regime of the soil: the nitrogen absorption by plant roots sharply decreases at temperatures below 5–6°C [16]. As a result of this fall, plant organic matter replenishes the supersoil layer of organic residues, i.e., the supersoil layer of detritus. This process returns the nitrogen of plants into the soil and includes it into microbiological processes. Annually, 17–25 gN/m² are returned with the fall into the soil in moist tropical

forests, in subtropical forests this return is 12–23 gN/m², in temperate-zone forests it is 3–8 gN/m², in steppes it is 4–9 gN/m², in deserts it is 2–4 gN/m², and in tundra it is 1–2 gN/m² [8].

Organic matter in the supersoil layer of detritus decomposes through two processes: *mineralization* and *humification*. It should be noted that most of the organic matter of plant residues is mineralized, whereas its smaller part is transformed as a result of humification. In the process of humification, nitrogen of the supersoil layer of detritus replenishes the content of organic compounds of soil nitrogen. The mineralization of nitrogen in the supersoil layer of detritus in soils transfers nitrogen-containing organic compounds into inorganic compounds through ammonification and nitrification. *Ammonification* is the microbiological transformation of nitrogen of organic compounds into ammonium ion NH_4^+ or ammonia NH_3 . This process proceeds under anaerobic conditions (in the absence of oxygen). A certain part of the formed nitrogen compounds is absorbed by the root systems of plants (*nitrogen assimilation, uptake by plants*). *Nitrification* is the process (proceeding under aerobic conditions) of the bacterial oxidation of ammonia to form the nitrite ion NO_2^- by some bacteria and then into the nitrate ion NO_3^- by another group of bacteria. The intensity of decomposition of organic matter depends substantially on thermal and oxidation–reduction conditions. At low temperatures, nitrification proceeds very slowly; its rate grows as the temperature increases and attains its optimum at 34.5°C [16]. Thus-formed nitrates can be absorbed by plants and soil–microbe populations. It should be noted that soil microorganisms, as a result of intense decomposition, can absorb and fix a considerable amount of accessible mineral nitrogen of the soil in their mass (*nitrogen immobilization*). This process, at a sufficiently high rate of nitrogen immobilization in the soil, often deteriorates the conditions of nitrogen nutrition of plants and increases the soil respiration [11]. It should be noted that nitrous oxide is formed during nitrification as an intermediate product and can be released into the atmosphere.

The process of denitrification caused by some bacteria decreases the resources of nitrate nitrogen in the soil. As a result of denitrification, nitrates are reduced to gaseous molecular nitrogen released into the atmosphere. Nitrogen oxides NO and NO_2 are formed as intermediate products in the process of denitrification and are also supplied into the atmosphere. The intensity of denitrification depends on the temperature, humidity, soil acidity, and soil pollution. The annual production of nitrogen as a result of the denitrification process on the land is estimated from 13–20 to 350–400 MtN/year [5, 9].

In the aboveground part of the cycle, nitrogen is supplied into sediments, for example, as a result of the potassium nitrate formation and leaching. This flux is

estimated at 0.2 MtN/year [27]. As a result of the peat-formation process, about 20 MtN/year are removed from the biological turnover. Due to fires in natural ecosystems, from 10 to 200 Mt of nitrogen are annually supplied into the atmosphere.

A certain part of nitrogen is supplied from continents into the ocean with river and underground discharges. The intensity of this flux is estimated at 24–80 MtN/year [5, 16, 28]. Annually, 9–16 Mt of inorganic nitrogen with the river discharge, 9–41 Mt of nitrogen in the composition of dissolved organic matter, and 6–55 Mt of nitrogen with particulate organic matter are supplied into the ocean [5, 14, 17].

Nitrogen is present in the ocean in the form of various organic compounds (living biomass and dissolved and particulate organic matter) and as dissolved molecular nitrogen, ammonium ions, nitrite, and nitrate. According to [5, 9, 14, 15, 23], 10–22 Tt of molecular nitrogen is dissolved in the world ocean. The nitrogen resources in the ocean in the composition of detritus are 45–900 GtN and in the composition of biomass of plants and animals, 200–800 MtN. The nitrogen content in the form of inorganic compounds in the ocean is estimated at 99–690 GtN. The living biomass of the ocean is concentrated mainly in its near-surface layers.

The nitrogen resources in the ocean are replenished due to atmospheric fixation (30–82 MtN/year) and the water discharge from the land (24–80 MtN/year) [5, 16, 17]. The annual biological fixation of atmospheric nitrogen in the water medium is estimated from 10–20 to 100–200 MtN [9, 23, 28, 29]. In the ocean, the largest part of detritus (more than 90%) is decomposed in its surface layers [30]. A lesser part of detritus is transported into deep layers of the ocean in the process of gravitational deposition. In deep layers of the ocean, detritus undergoes decomposition and an insignificant amount of it passes into bottom sediments; i.e., it is removed from the biological turnover. According to [5, 28], from 1 to 16 Mt of nitrogen are annually supplied into oceanic sediments in the composition of organic matter. Inorganic compounds of nitrogen formed in deep layers rise as a result of turbulent diffusion and vertical advection into upper layers of the ocean. The considerable nitrogen emission into the atmosphere is associated with the process of denitrification in the ocean. This emission is estimated from 25 to 454 MtN/year [23, 28]. An insignificant part of oceanic nitrogen (to 1 MtN/year [31]) is supplied to the land surface as a result of fishery.

Anthropogenic impacts considerably affect the current global nitrogen cycle (Table 3). The production and application of mineral nitrogenous fertilizers introduce substantial changes into the nitrogen cycle. These fertilizers supply about 90 Mt of nitrogen annually into terrestrial ecosystems [32]. The biota utilizes no more than 50% of mineral nitrogen introduced into the soil [33]. An insignificant amount (2–5%) is accumulated in soils. A considerable fraction of mineral

Table 3. Nitrogen fluxes in the biosphere caused by anthropogenic impact [5, 27, 28, 32, 33]

Flux	(MtN/year) value
Biological fixation	25–90
Denitrification	17–20
River runoff from land into ocean	20–35
Supply into groundwater	12–15
Emissions into the atmosphere	40–60
Industrial fixation	60–125
Supply into soil with fertilizers	80–90

Table 4. Estimates of the main nitrogen fluxes in the biosphere [5, 14–18, 23, 27, 28]

Flux	(MtN/year) value
Atmospheric fixation:	
over land	20–63
over ocean	23–82
Biological fixation:	
on land	30–290
in ocean	10–200
Denitrification:	
on land	13–400
in ocean	25–454
Turnover “photosynthesis ↔ destruction of organic matter”:	
in terrestrial ecosystems	1200–3500
in ocean	4000–6000
Sink from land into ocean:	24–80
dissolved organic matter	8.6–41
particulate organic matter	6.1–55
dissolved inorganic compounds	9.2–16
Supply into sediments:	
on land	20
in ocean	1–16
Natural sources of geosphere	1–9

nitrogen introduced into the soil volatilize in the form of gaseous compounds forming during the transformation of nitrogen contained in fertilizers (up to 25%) and is washed out into groundwater and open bodies of water (up to 20%), enhancing the greenhouse effect and worsening the biochemical state of the water medium. The nitrogen amount supplied into the atmosphere as a result of mineral fuel combustion and emissions of industrial enterprises and traffic is 40–60 MtN/year [5, 32, 33].

It should be noted that some links of the global nitrogen cycle on the land and in the water medium are insufficiently investigated. Some quantitative estimates of the nitrogen contents in natural components and the intensities of nitrogen fluxes possess a high degree of uncertainty. Microbiological processes, both in the soil and in the water medium, are least investigated in the biochemical nitrogen turnover. Uncertainty ranges of estimates of nitrogen fluxes in the biosphere are given in Tables 4 and 5.

MODELING OF THE NITROGEN CYCLE

The nitrogen turnover has been studied for more than two centuries (see Fig. 3). Gaseous nitrogen called “suffocating air” has been known since the 1870s. It has been known that nitrogen is one of the basic elements in living organisms, it has a substantial influence on the development of plants, and its compounds are effective fertilizers since the middle of the 19th century [28]. Processes of biological nitrogen fixation by soil microorganisms, as well as processes of nitrification and denitrification, were investigated in the beginning of the 20th century.

Vast data on local components of the global nitrogen cycle, on the role of nitrogen in biospheric processes, and on the anthropogenic impact on the nitrogen turnover were accumulated by the 1970s. These data were analyzed and generalized in [23]. A series of conceptual balance models of exchange processes in some natural ecosystems were constructed on the basis of empirical data [35, 36]. These models considered the dynamics of resources of carbon, nitrogen, and a set of mineral elements in the atmosphere, vegetation, and soil, as well as the intensities of fluxes of these substances. An analysis of the models showed that the ecosystems under consideration are characterized by the absence of balance for annual biogeochemical cycles of some chemical elements, including nitrogen. The noted unbalance of production and destruction processes in local ecosystems can be the driving force of evolution of both natural ecosystems and the biosphere as a whole [36].

In order to predict the consequences of anthropogenic impacts on the biosphere and individual plant communities, simulation models describing of various chemical elements were developed. A model of biogeochemical processes of carbon, nitrogen, and phosphorus is described in [37]. Estimates of the action of toxic chemicals on the cycles of the considered elements on the land and in the ocean were obtained on the basis of this model. The dynamics of nitrogen in phytomass, detritus, and the soil of a local grass ecosystem at different levels of introduction of nitrogenous fertilizers were analyzed in [38] with the use of a mathematical model. The first, rather simplified simulation models demonstrated that the turnover of biogenic elements can be, in principle, investigated by formal mathematical methods.

At that time, for the first time in the practice of mathematical modeling, the development of large-scale models of the interaction between biogeochemical, climatic, and socio-economic processes started at the Computational Center of the Academy of Sciences of the USSR in order to create a simulation system of models and predict the interaction between society and the environment on their basis. The first version of the global model of the biosphere (the Moscow biospheric model (MBM)) was developed in the early 1980s [39]. This model consisted of the biospheric, climatic, and demographic blocks, as well as the block of anthropogenic impacts on the biosphere. The biospheric block included models of the global turnovers of carbon, oxygen, nitrogen, and water; models of oceanic biota; and models of terrestrial ecosystems. The structure of the model can serve as an example for the development and introduction of additional blocks describing the turnovers of chemical elements into modern climatic models.

One of the first models of the global nitrogen cycle interacting with the global carbon cycle in the atmosphere–plants–soil system was developed within the framework of the MBM. Six components—nitrogen of the atmosphere, nitrogen of plants, nitrogen of the supersoil layer of detritus, nitrogen of organic compounds of the soil, nitrogen of mineral compounds of the soil, and exported nitrogen—are considered in the nitrogen turnover scheme. The intensities of nitrogen fluxes between these components depend on the nitrogen contents in them and the C : N ratios in the corresponding components. The role of nitrogenous nutrition of plants in the response of the atmosphere–plants–soil system to external actions was analyzed with the use of the proposed model of the nitrogen cycle. In particular, the response of the system under consideration to an instantaneous increase in the carbon dioxide content in the atmosphere (from 10 to 100%) was investigated at different degrees of dependence of the vegetation-cover productivity on the nitrogenous nutrition. Imitative experiments showed that the stronger the dependence of the productivity of plants on the amount of assimilated nitrogen in the soil is, the more rapidly the carbon dioxide emission is absorbed by terrestrial ecosystems [39].

In works [40–42], the combined model of the nitrogen and carbon cycles in the atmosphere–plants–soil system is supplemented with the model of interaction of the cycles of these elements in the atmosphere–ocean system. The ocean is divided into two layers in depth: surface and deep. Three reservoirs of carbon and nitrogen—living biomass, detritus, and dissolved inorganic compounds—are identified in the surface layer of the ocean. The deep layer of the ocean consists of two reservoirs of nitrogen and carbon compounds: detritus and dissolved inorganic compounds. The response of the global atmosphere–land ecosystems–ocean system to instantaneous emissions of carbon dioxide into the atmosphere is investigated on the

Table 5. Components of the nitrogen balance in the biosphere [5, 6, 14, 15, 18, 23, 28, 34]

Flux	(MtN/year) value
Supply from atmosphere:	
into terrestrial ecosystems	
N_2	140–260
NH_x	34.1–186
NO_y	24.8–83
into ocean	
N_2	30–130
NH_x	12.4–50
NO_y	11–33
Return into atmosphere:	
from terrestrial ecosystems	
N_2	64.2–92
N_2O	4.2–69
NH_x	57–244
NO_y	13–108
from the ocean	
N_2	5–450
N_2O	2–80
NH_3	3–13
Supply from land into ocean:	
NH_x	30–43
NO_y	5–11

$$NH_x = NH_3 + NH_4^+.$$

NO_y = any N–O compounds except N_2O : NO_x , NO_3 , N_2O_5 , HNO_3 , $HONO$ etc.

basis of the developed model. According to the results of numerical experiments with the model, the inclusion of the nitrogen cycle substantially affects the distribution of carbon between the atmosphere, the terrestrial biota, and the ocean [40].

A global model for the analysis of the spatial distributions of the characteristics of dynamic processes in the biosphere is proposed in [27, 43]. The proposed simulation model includes the following blocks: the model of terrestrial plants; the model of functioning of the oceanic ecosystem; and the model of global biogeochemical cycles of carbon, nitrogen, phosphorus, and sulfur. The model of global nitrogen cycle is realized in the atmosphere–terrestrial ecosystems–ocean system. In this model, nitrogen compounds in the atmosphere form a unified generalized nitrogen reservoir. The ocean is divided into four layers in depth: the photic layer with illumination sufficient for the process of photosynthesis, the intermediate layer (which is also photic), the deep layer (without photosynthesis), and the near-bottom layer as a boundary layer.

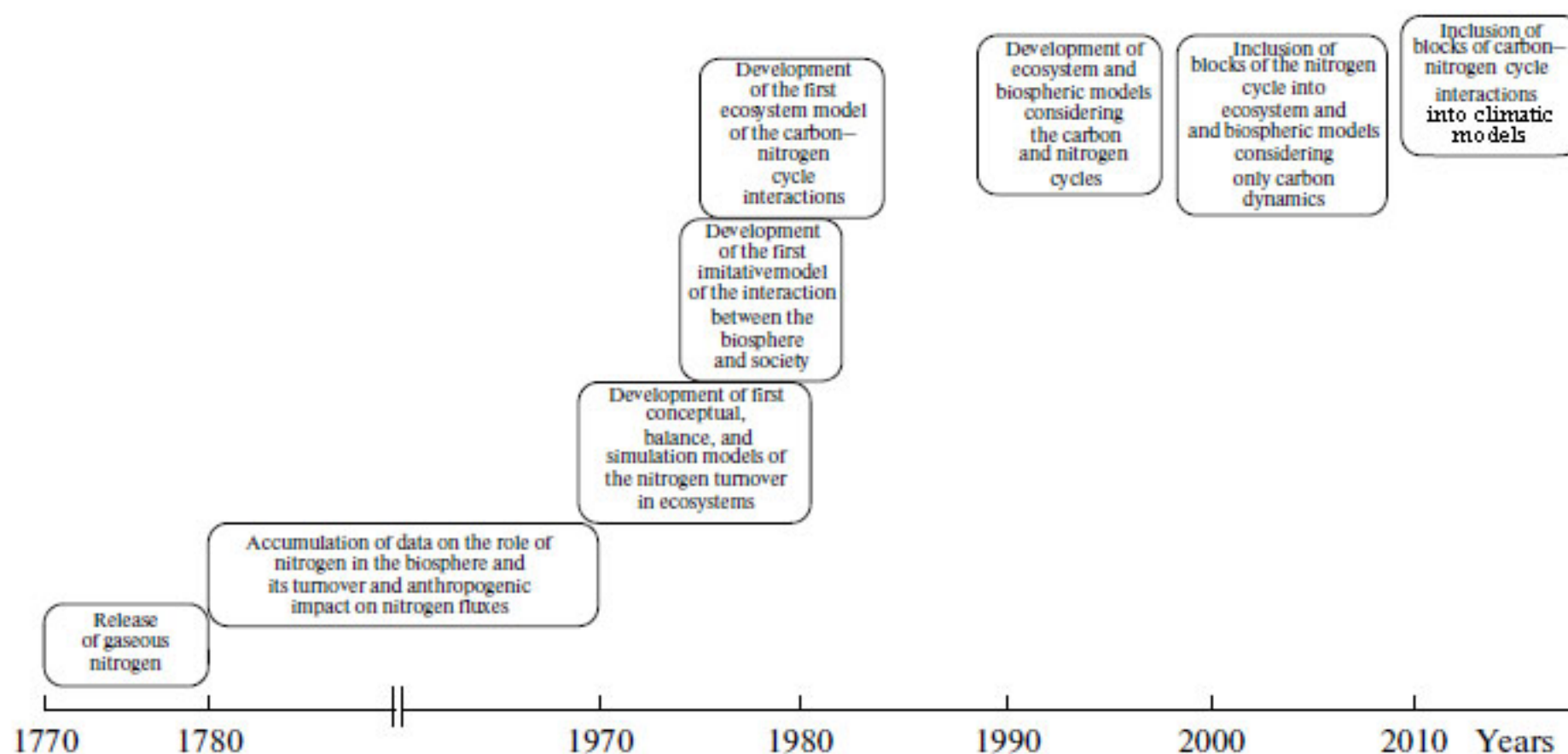


Fig. 3. Basic periods in the study of the biogeochemical nitrogen cycle.

Terrestrial ecosystems are considered sets of two nitrogen reservoirs: nitrogen in living organic matter and the nitrogen of the soil assimilated by plants. The proposed model of the biosphere makes it possible to calculate the dynamics of various biospheric components under large-scale actions.

The CYCLES model [17] was developed at the beginning of the 1990s for a system analysis of feedbacks between the biogeochemical cycles of carbon, nitrogen, phosphorus, and sulfur, as well as for studying the biospheric turnovers of these elements. The global nitrogen cycle model of the CYCLES model simulates the dynamics of N_2 , NH_3/NH_4^+ , N_2O , NO , NO_2 , NO_2^- , and NO_3^- in the atmosphere–terrestrial ecosystems–ocean system. This model consists of the ocean box diffusion model, the model of terrestrial ecosystems (which includes the fresh water system: ground water and fresh surface water), and the atmospheric chemistry model. According to the results of model calculations, the inclusion of an interaction between the carbon and nitrogen cycles has a substantial influence on the estimation of the carbon budget of terrestrial ecosystems. The deficit of mineral forms of soil nitrogen considerably decreases the consumption of carbon dioxide from the atmosphere by vegetation.

The conceptual scheme of a model of the nitrogen turnover in the biosphere, as well as functional dependences of the main nitrogen fluxes between the atmosphere, terrestrial ecosystems, and the ocean, are proposed in [44] (see Fig. 4). In this model, nitrogen of the atmosphere is presented by the universal reservoir connecting the nitrogen cycles on the land and in the ocean. Such nitrogen reservoirs as vegetation, the supersoil layer of detritus, soil inorganic (mineral) compounds, and soil organic compounds are also

identified in terrestrial ecosystems. Taking into account that the bulk of the mass of oceanic living organic matter is concentrated in the surface ocean layer up to 200 m deep, the ocean is presented as two nitrogen reservoirs: the upper ocean layer and the deep ocean.

Processes of exchange by nitrogen compounds take place between the ocean layers under consideration. The nitrogen content in vegetation is replenished due to biological fixation and assimilation of inorganic nitrogen of the soil by vegetation. Nitrogen is supplied into the supersoil layer of detritus with the fall of vegetation organic matter, and this layer is mineralized and humified under the action of microorganisms. These processes replenish soil reservoirs of inorganic and organic compounds of nitrogen, respectively. The mineralization of organic matter of the soil, atmospheric fixation on the land, and the introduction of fertilizers into the soil replenish resources of inorganic compounds of soil nitrogen. As a result of denitrification, the nitrogen of inorganic compounds of the soil is released into the atmosphere. A certain amount of inorganic compounds of soil nitrogen is supplied into the surface ocean layer with the continental runoff. The surface ocean layer also becomes enriched in nitrogen through atmospheric fixation. The denitrification process decreases the nitrogen resources in the ocean. Detritus decomposes both in the surface and deep ocean layers. A certain amount of nitrogen is supplied into bottom sediments in the composition of organic matter. The supply of juvenile nitrogen and anthropogenic emissions of nitrogen compounds replenish the atmospheric reservoir.

A model of the global turnovers of carbon and nitrogen in the atmosphere–ocean system, which takes into account the spatial structure of the ocean

and reproduces the seasonal dynamics, was developed in order to study the spatial dynamics of carbon dioxide between the atmosphere and the ocean and to investigate the role of the ocean in the process of CO₂ absorption and stabilization of the global climate [42]. In this model, the ocean is divided in depth into three layers: upper, thermocline, and deep. The atmosphere in the model is assumed to be homogeneous over height, and it is presented by a single layer. This model was used both for calculating the global cycles of nitrogen and carbon under undisturbed conditions and for investigating the consequences of anthropogenic loads on the biosphere.

By now, models are developed for investigating nitrogen fluxes in a number of marine ecosystems [45, 46]. A three-dimensional ecosystem model with a hydrodynamic block which takes into account the interrelation of biogeochemical cycles of carbon and nitrogen is proposed in [47]. This model is used for a more exact estimation of the annual nitrogen flux in the process of denitrification from the northwestern territory of the Indian Ocean. The factors responsible for the spatial and temporal distributions of the denitrification intensity were revealed on the basis of imitative experiments. The NEMURO model [48] was developed for a study of the seasonal and interannual dynamics of the biogeochemical cycles of nitrogen, carbon, and silicon in the Pacific Ocean. This model was used to estimate the response of ecosystems in different regions of the Pacific to climate warming.

THE NITROGEN CYCLE IN MODELS OF TERRESTRIAL ECOSYSTEMS

The results of numerous field investigations indicate that the interaction between the carbon and nitrogen cycles in terrestrial ecosystems has a substantial influence on changes in the nitrogen and carbon contents in vegetation, the supersoil layer of detritus, and the organic matter of the soil [8, 11, 49]. Dynamic models of the combined cycles of carbon and nitrogen in concrete ecosystems are developed in order to estimate the tendencies of development of local ecosystems and the factors controlling the dynamics of their functioning. A dynamic model of the combined turnover of organic matter and nitrogen in a bog ecosystem [50] was developed at the beginning of the 1980s. The role of biogenic and hydrologic factors in the dynamics of the system under investigation was elucidated on the basis of model calculations. The boundaries of homeostasis and the conditions under which the studied system loses its stability were investigated with the use of this model. Ecosystem models, which consider carbon–nitrogen cycle interactions, started to be actively developed by the beginning of the 1990s. The first versions of models such as TEM [51], CENTURY [52], and CASA [53] were elaborated. It should be noted that the carbon turnover was considered in these models as the main cycle, whereas the nitrogen

cycle was more likely a subordinate cycle, which corrected the carbon cycle.

The increasing content in the atmosphere of greenhouse gases, among which carbon dioxide plays the main role, is a key factor of present-day climate changes [54]. In recent decades, most attention has been concentrated on the problems of the carbon turnover, such as the study of natural carbon sources, sinks, and reservoirs; the estimation of intensities of the production and destruction processes in ecosystems; making carbon balances at the regional and global levels; and predicting the expected response of carbon-flux velocities to probable climate changes. Physicomathematical models of the carbon cycle (see, for example, [43, 55–58]) developed for solving these problems on various temporal and spatial scales did not take into account the influence of turnovers of other biogenic elements on this cycle. It should be noted that, despite the interrelation of the biospheric cycles of carbon and nitrogen established from experimental and imitative investigations [11, 40, 49], most modern biospheric models disregard this interrelation in modeling the carbon dynamics. Some biospheric models that were developed in order to analyze the intensities of fluxes of the carbon cycle under different scenarios of anthropogenic impacts were supplemented with the nitrogen cycle (for example, the ORCHIDEE [59], LPJ [60], ISAM [61], and UVicESCM [62] models).

In modern biospheric models, terrestrial ecosystems are usually presented as four blocks: vegetation, the supersoil layer of detritus, organic compounds of the soil, and inorganic compounds of the soil. In some models, for example, in the CN-TEM model [63], the supersoil layer of detritus is considered a part of the organic compounds of the soil. In different models, each of the four model blocks is described with one or other degree of detail. For example, in the GNCC model [24], vegetation is presented as a unified nitrogen block, whereas organic and inorganic nitrogen compounds of the soil include two nitrogen subblocks each; in the O-CN model [59], vegetation is presented in nine subblocks; the supersoil layer of detritus is presented in six subblocks; and soil organics are presented in four subblocks. In these models, the dynamics of nitric gaseous compounds in the atmosphere is excluded from consideration. The intensities of nitrogen fluxes between model components depend on the contents of nitrogen and carbon in them and on a number of biophysical parameters.

Biospheric models were used for investigating the influence of the nitrogen cycle on parameters of the terrestrial carbon cycle with a change in the atmospheric CO₂ concentration, climatic conditions, and the land-use character. The inclusion of nitrogen dynamics substantially corrects estimates of the spatial distribution and spatial dynamics of terrestrial sources and sinks of carbon, which were obtained with the use of models taking into account the carbon cycle alone

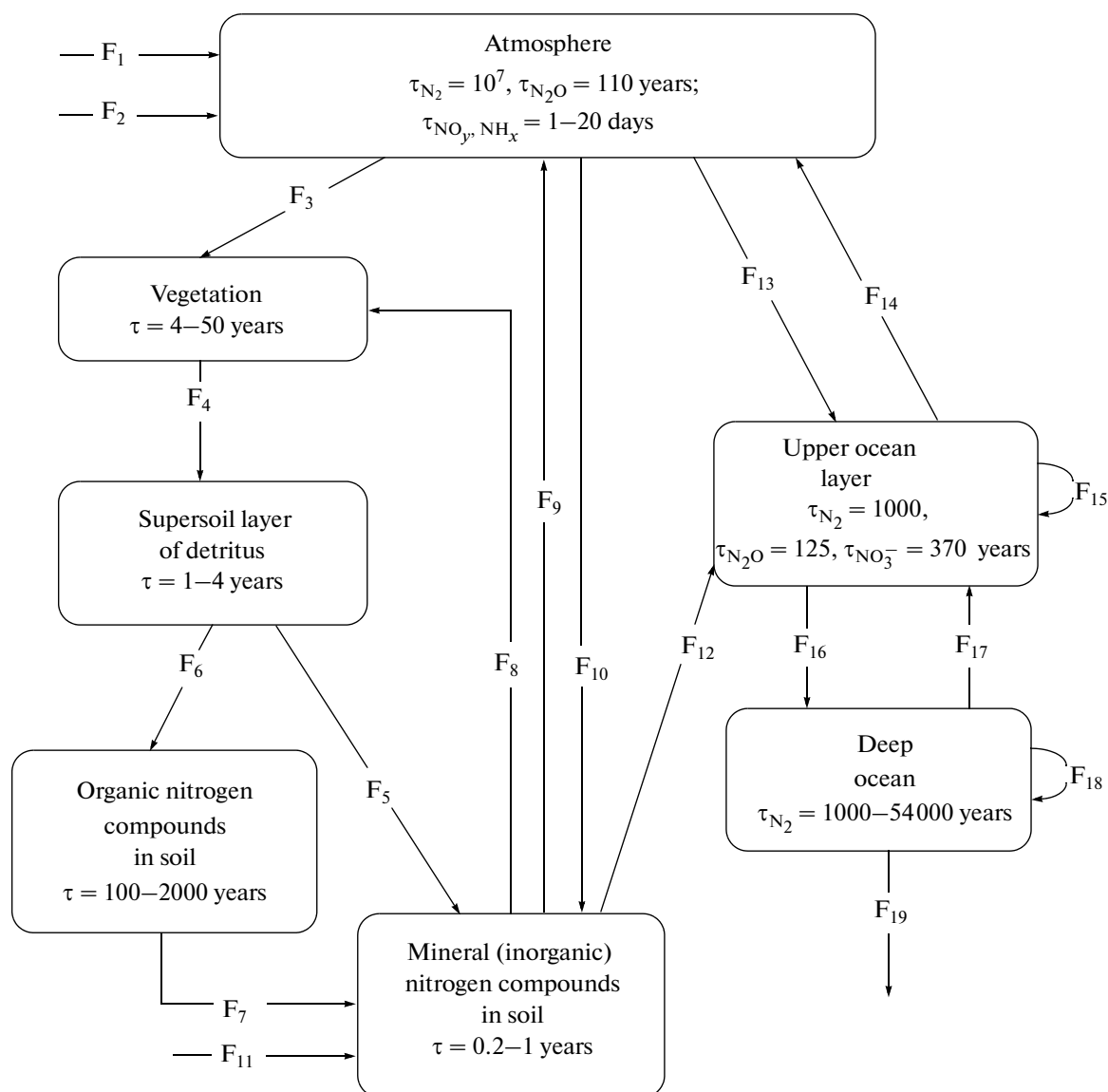


Fig. 4. Conceptual scheme of the model of nitrogen turnover in the biosphere: (F_1) supply from natural sources of the geosphere, (F_2) anthropogenic emissions into the atmosphere, (F_3) biological fixation on the land, (F_4) litterfall, (F_5) mineralization of detritus of the supersoil layer, (F_6) humification of the detritus of the supersoil layer, (F_7) mineralization of soil organic matter, (F_8) uptake by plants, (F_9) denitrification from the land, (F_{10}) atmospheric fixation on the land, (F_{11}) supply with fertilizers, (F_{12}) river discharge, (F_{13}) atmospheric fixation over the ocean, (F_{14}) denitrification from the ocean, (F_{15}) decomposition of detritus and biological fixation in the ocean, (F_{16}) supply of detritus from upper layers into the deep ocean, (F_{17}) rise of inorganic nitrogen compounds into upper layers of the ocean, (F_{18}) decomposition of detritus in the deep ocean, (F_{19}) sedimentation, and (τ) the residence time of nitrogen in components of the biosphere according to [8, 14, 23].

[59–67]. In particular, for regions where mineral compounds of soil nitrogen are a limiting factor for the development of vegetation cover, the productivities of the vegetation cover and carbon resources in the soil were more adequate for experimental data.

The inclusion of the combined dynamics of the carbon and nitrogen cycles into models decreases the intensity of carbon dioxide absorption by the vegetation cover when the CO_2 concentration in the atmosphere increases and the near-surface temperature remains constant [64, 65]. This effect is mainly caused

by a deficit of soil nitrogen accessible for plants. The deficit of mineral nitrogen in the soil decreases both the productivity of vegetation cover and the decomposition rate of supersoil detritus and organic matter of the soil in many ecosystems, in particular, in boreal forests [8].

Model estimates indicate that, in some ecosystems, resources of mineral nitrogen in the soil can decrease if the atmospheric CO_2 concentration increases [66]. Global estimates obtained with the use of biospheric models taking into account the carbon–nitrogen

interaction indicate that carbon resources in terrestrial ecosystems can increase during an increase in the atmospheric CO_2 concentration and climate warming [65, 67].

THE NITROGEN CYCLE AND CLIMATIC MODELS

In recent years, climatic models have been supplemented with blocks of the carbon cycle [68–71]. The results of numerical calculations with these models showed that the climate warming associated with an increase in the atmospheric carbon dioxide concentration leads to a decrease in the carbon dioxide deposition in terrestrial ecosystems. This effect is caused by increases in the atmospheric carbon concentration and near-surface temperature, which favor the intensification of both the production and destruction processes in ecosystems. Changes in the intensities of these processes increase the decomposition rate of soil organic matter, soil respiration (i.e., the carbon dioxide emission from the soil into the atmosphere), and the sensitivity of vegetation productivity to the soil moisture and air temperature. According to the results, by the end of the 21st century, the soil respiration intensity in some terrestrial ecosystems may exceed the rate of accumulation of atmospheric carbon by the vegetation cover. In this case the ecosystems can transform from a sink of carbon from the atmosphere into its source. The results of numerical experiments with climatic models incorporating the carbon cycle revealed a positive feedback between climate and the carbon cycle [72–74]. It should be noted that the globally averaged near-surface temperature serves as an integral characteristic of climate, whereas the carbon dioxide concentration in the atmosphere is an integral characteristic of the carbon cycle.

To analyze the influence of the nitrogen cycle on the carbon exchange between components of the climatic system, some climatic models were supplemented with blocks of interaction between the carbon and nitrogen cycles [63, 75]. The calculations showed that the inclusion of the nitrogen cycle into the terrestrial biogeochemical block of a climatic model substantially corrects the model results obtained previously. In particular, a joint consideration of the carbon and nitrogen cycles in a climatic model decreases the carbon sink into terrestrial ecosystems caused by the increase in the atmospheric carbon dioxide concentration by 53–78% compared with the results obtained with the use of models taking into account the carbon cycle alone [63, 76].

Based on calculations with the use of the general circulation model of the atmosphere and ocean (CCSM), it is shown in [75, 76] that the CO_2 stimulation (fertilization) effect (the productivity increase of the vegetation cover due to increasing carbon dioxide absorption by it upon the atmospheric CO_2 concentration increase) is considerably smaller with the inclu-

sion of the interaction between the carbon and nitrogen cycles (CN model) than when only the carbon turnover is considered (C model). Calculations show that the total carbon absorption by vegetation due to the atmospheric CO_2 concentration increase for the period 2000–2100 in accordance with the SRES A2 scenario for the C model is by a factor of 3.8 higher than for the CN model [76]. The results of calculations with the CN model showed that, in the absence of a near-surface temperature increase, the deficit of mineral forms of soil nitrogen is mainly responsible for lowering the CO_2 stimulation effect. According to the results in [75], the incorporation of the nitrogen cycle into a climatic model substantially decreases the response of the terrestrial carbon cycle to the interannual variability of the near-surface temperature and amount of precipitation. In the cited work, the response of the terrestrial carbon cycle is determined as a change in the net carbon exchange in ecosystems (balance between the supply of carbon into the ecosystem as a result of the photosynthesizing activity of the vegetation cover and the carbon release as a result of autotrophic respiration, processes of organic matter destruction, and fires).

The results of investigating the nitrogen cycle influence in terrestrial ecosystems on the interrelation between the carbon cycle and climate obtained on the basis of the IGSM climatic model of intermediate complexity are presented in [63]. The results of modeling showed that the inclusion of the carbon–nitrogen interrelation not only limits the CO_2 stimulation effect, but also can change the sign of the feedback between climate and the carbon cycle. In the absence of carbon–nitrogen interrelation, global warming decreases resources of terrestrial carbon both in the vegetation cover and in the soil as a result of the growth in autotrophic respiration and the decomposition rate of organic matter in the supersoil layer of detritus and in the soil; i.e., the carbon–climate feedback is positive. If the nitrogen dynamics is taken into account in the model, the estimates obtained indicate that the resources of terrestrial carbon grow at a near-surface temperature increase caused by increasing CO_2 concentration in the atmosphere. This is due to the fact that an increase in the decomposition rate of organic matter leads to the accumulation in the soil of mineral nitrogen accessible for plants. Increasing resources of mineral nitrogen compounds in the soil increase the productivity of terrestrial plants, i.e., enhance the so-called *N stimulation effect*. Increasing the rate of soil–nitrogen mineralization stimulates the growth of plants, i.e., increases the absorption of atmospheric carbon by the terrestrial vegetation cover. First and foremost, the N stimulation effect will manifest itself in ecosystems for which mineral compounds of soil nitrogen are a limiting factor for the vegetation-cover development, for example, in boreal and mixed forests [77].

Model calculations with allowance for the dynamics of carbon and nitrogen in a climatic model [63] showed that, during climate warming, the consumption of atmospheric carbon by the intensely developing vegetation cover not only will compensate for soil-carbon emission, but can even exceed it. For this reason, as a result of warming, terrestrial ecosystems will be a carbon sink. Therefore, if the carbon–nitrogen dynamics are considered, global warming can favor the accumulation of carbon resources in terrestrial ecosystems; i.e., the climate–carbon cycle feedback can become negative. Work [76] also points to a negative feedback between climate and the carbon cycle, when interactions of the carbon and nitrogen turnovers are considered in a climatic model. According to [76], such a negative feedback is caused by the intensification of the growth rate of vegetation due to the increasing mineralization of soil nitrogen during climatic warming.

The scheme of calculation of the coefficients characterizing the carbon dioxide stimulation effect on terrestrial plant cover and the interaction between climate and the carbon cycle is proposed in [69]. Dependence

$$\Delta C(t) = \beta \Delta C_A(t) + \gamma \Delta T(t),$$

is proposed for the calculation of carbon-content changes in terrestrial ecosystems $\Delta C(t)$ in the linear approximation. Here, $\Delta C_A(t)$ is a change in the concentration of carbon dioxide in the atmosphere, ΔT is a corresponding change in the globally averaged near-surface air temperature; β and γ are diagnostic coefficients. The coefficient β characterizes the carbon dioxide stimulation effect, and the coefficient γ characterizes the interaction of climate and the carbon cycle.

Two numerical experiments under the same scenario of anthropogenic CO_2 emissions into the atmosphere and the same initial conditions were performed with a climatic model in which the carbon cycle was realized. A complete climatic model was used in the first experiment. The second experiment used a model with a unidirectional relation between climate and the carbon cycle at which climate changes do not affect the $\Delta C(t)$ value. The values of the coefficient β are found from the $\Delta C_A(t)$ and $\Delta C(t)$ data (for example, through a linear regression), which are obtained in the second numerical experiment. It is supposed that the climate change is fairly small and, consequently, the coefficient β does not depend on time. The identity of initial conditions in calculations with the complete model and the model with a unidirectional relation under the assumption that the climate change is rather small makes it possible to use the calculated value of the coefficient β to analyze the results of calculations with the complete model. Therefore, the value of the coefficient γ can be determined from the presented dependence for the calculation of $\Delta C(t)$ and the results

obtained in calculations with the complete climatic model.

The case $\gamma < 0$ corresponds to a positive feedback between climate and the carbon cycle, which was revealed in climatic models with the carbon cycle (without regard for the nitrogen cycle) [71–73]. The growth of the carbon dioxide content in the atmosphere and the climate warming associated with it decreased the CO_2 absorption by terrestrial ecosystems and/or by the ocean, favoring an additional increase in the atmospheric carbon dioxide concentration. The interaction of the nitrogen and carbon cycles changes the values of the coefficient γ . On the whole, this interaction decreases the absolute value of γ , which points to the weakening of the feedback between climate and the carbon cycle. The inclusion of an interaction between the nitrogen and carbon cycles in [63] almost completely weakens the feedback between climate and the carbon cycle: the values of the coefficient γ remain negative but close to zero. It is shown in [76] that the value of this coefficient for terrestrial ecosystems can become positive, pointing to negative feedback between climate and the carbon cycle.

CONCLUSIONS

Stable interrelations and interdependences between global biogeochemical turnovers of chemical elements have been formed as a result of the evolution of the biosphere. In recent decades, human-related activities have become a substantial factor affecting the quantitative characteristics of the mass and energy exchange on the earth. Both the general structure of the biosphere and climatic processes have been subjected to the considerable influence of anthropogenic impacts. Mathematical models describing the dynamics of local and global biogeochemical processes in the biosphere and climatic changes associated with them, depending on one or other scenario of the anthropogenic impact on the environment, have been developed in order to obtain estimates of characteristics of such processes. At present there is a rather large number of conceptual and simulation models of the biogeochemical cycles of carbon, nitrogen, phosphorus, and sulfur [17, 18, 27, 43, 78–80]. Models of the nitrogen cycle were developed for estimating various actions on the dynamics of resources and fluxes of this element, both in the biosphere as a whole [17, 39, 40, 42] and in individual ecosystems [50, 81, 82]. These models are based on the block approach: The system under investigation is presented as a scheme consisting of several blocks and fluxes connecting them, which are described by functional dependences. The number of blocks in the system depends on the purpose of the investigation.

Nitrogen, being one of the main limiting factors for the development of vegetation cover and the intensity of destruction processes in ecosystems, has a substantial effect on the terrestrial carbon cycle [6, 8, 66, 83].

In recent years, modules describing the nitrogen cycle and its interrelation with the carbon cycle have been developed for some models of the carbon cycle [59–63]. According to calculations with the models which consider only the carbon cycle, the emission of carbon dioxide increases and its absorption by both terrestrial ecosystems and the ocean decreases during the climate warming caused by the accumulation of greenhouse gases in the atmosphere [72]. Calculations revealed a substantial influence of the nitrogen dynamics on the carbon-cycle response during the anthropogenic increase in the atmospheric carbon dioxide concentration and climatic changes [63, 66, 75]. According to the results of numerical experiments, the interaction of the carbon and nitrogen cycles in the biosphere substantially decreases the ability of terrestrial ecosystems to increase carbon absorption if the atmospheric carbon dioxide concentration increases and the near-surface temperature growth is absent, compared with the results obtained when the influence of the nitrogen cycle is ignored. According to calculations with the models considering carbon–nitrogen dynamics, the global warming caused by the growth of the atmospheric CO₂ concentration favors the accumulation of carbon resources in terrestrial ecosystems [59, 64, 65].

A positive feedback between climate and the carbon cycle has been established on the basis of climatic models considering only the biogeochemical cycle of carbon [71–74]. This positive feedback means that climate warming decreases carbon resources in terrestrial ecosystems (first and foremost, soil carbon resources). Numerical experiments with climatic models [63, 76] showed that including the interaction between the carbon and nitrogen cycles into these models weakens the positive feedback of climate and the carbon cycle up to the possibility of establishing a moderate negative feedback.

Therefore, accounting for the nitrogen cycle in climatic models can substantially change the estimate of the feedback between climate and the carbon cycle. Models with the combined dynamics of carbon and nitrogen make it possible to obtain more reliable estimates of the carbon exchange during expected climatic changes.

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